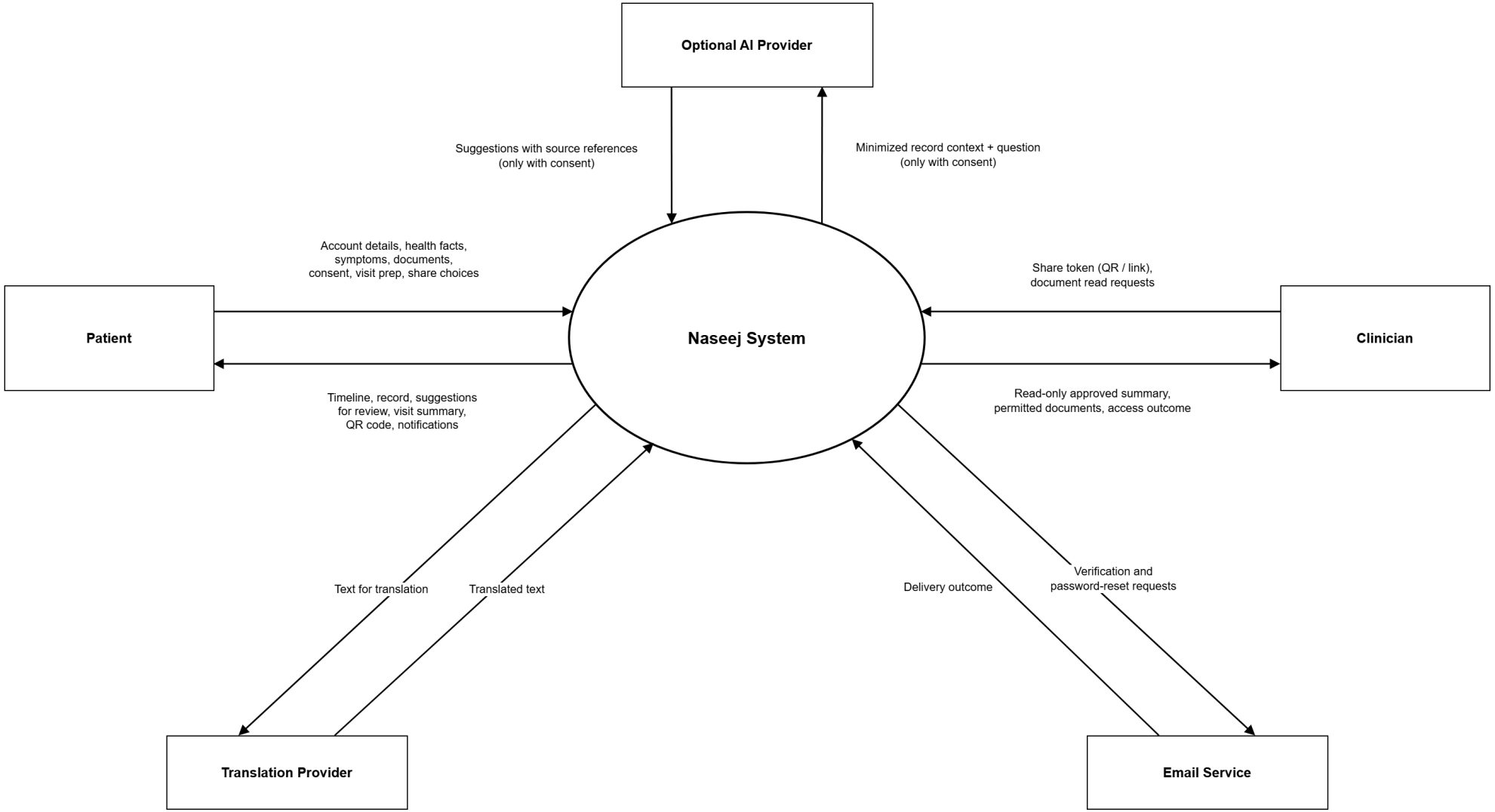
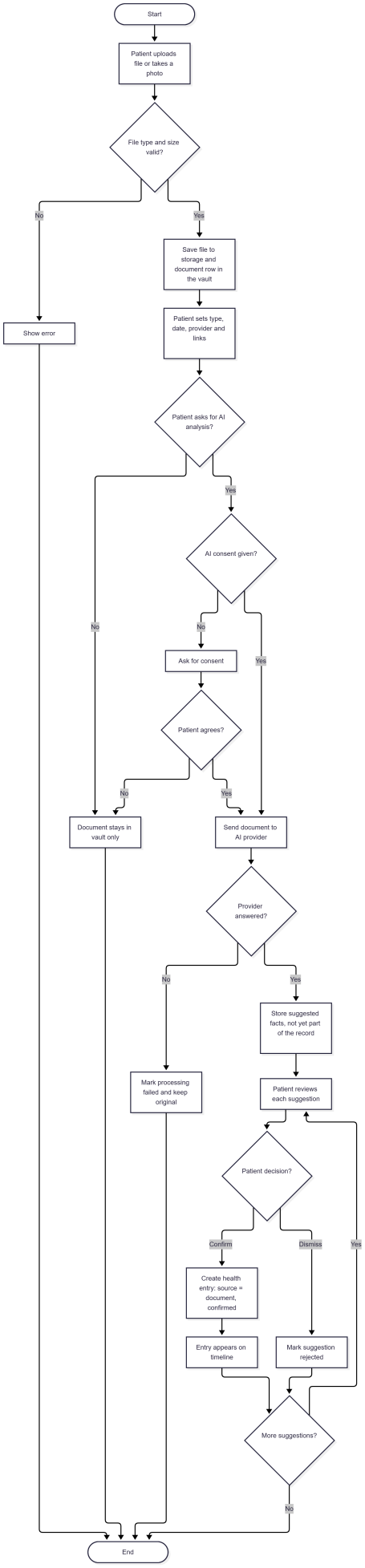
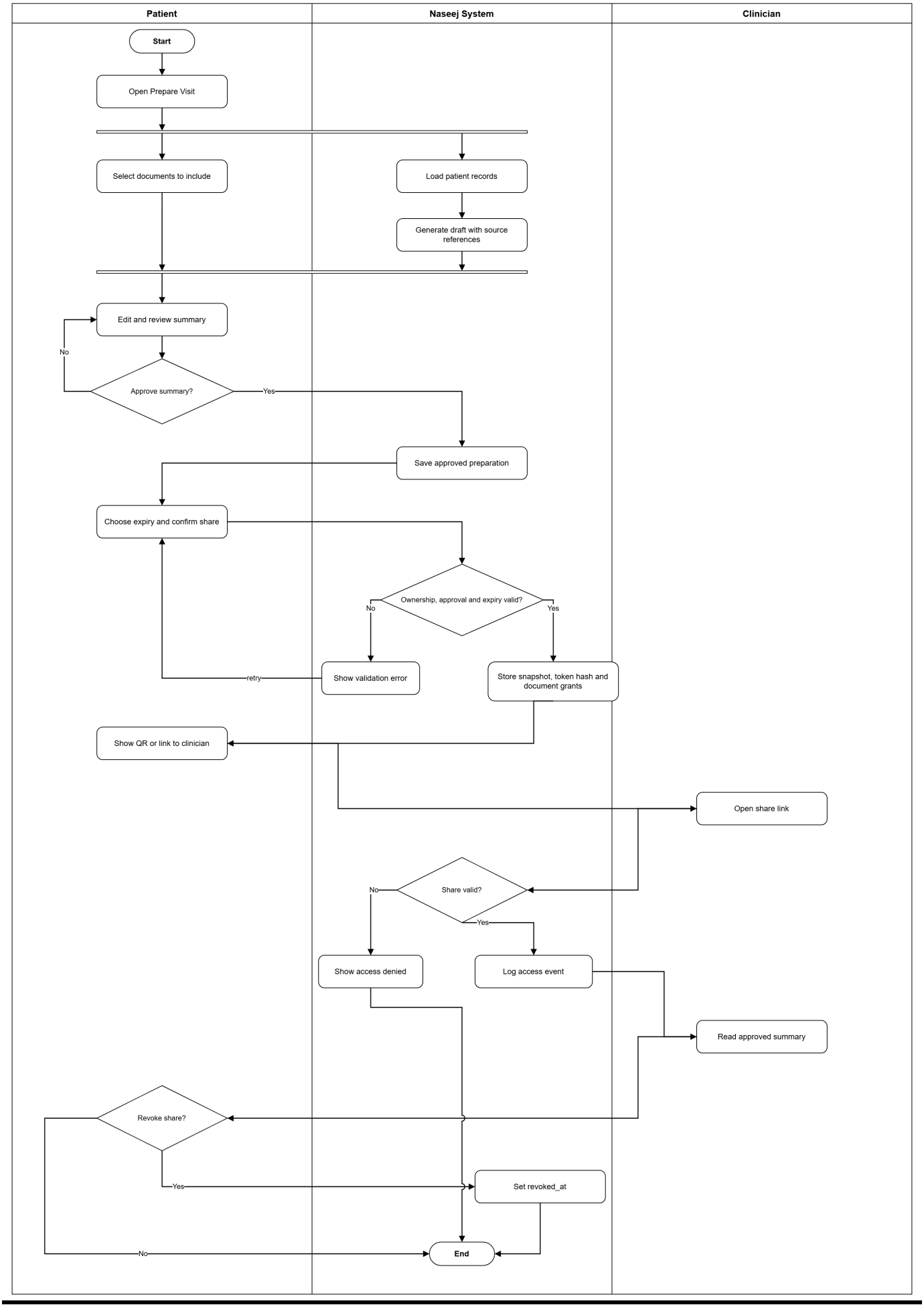
1. Context diagram



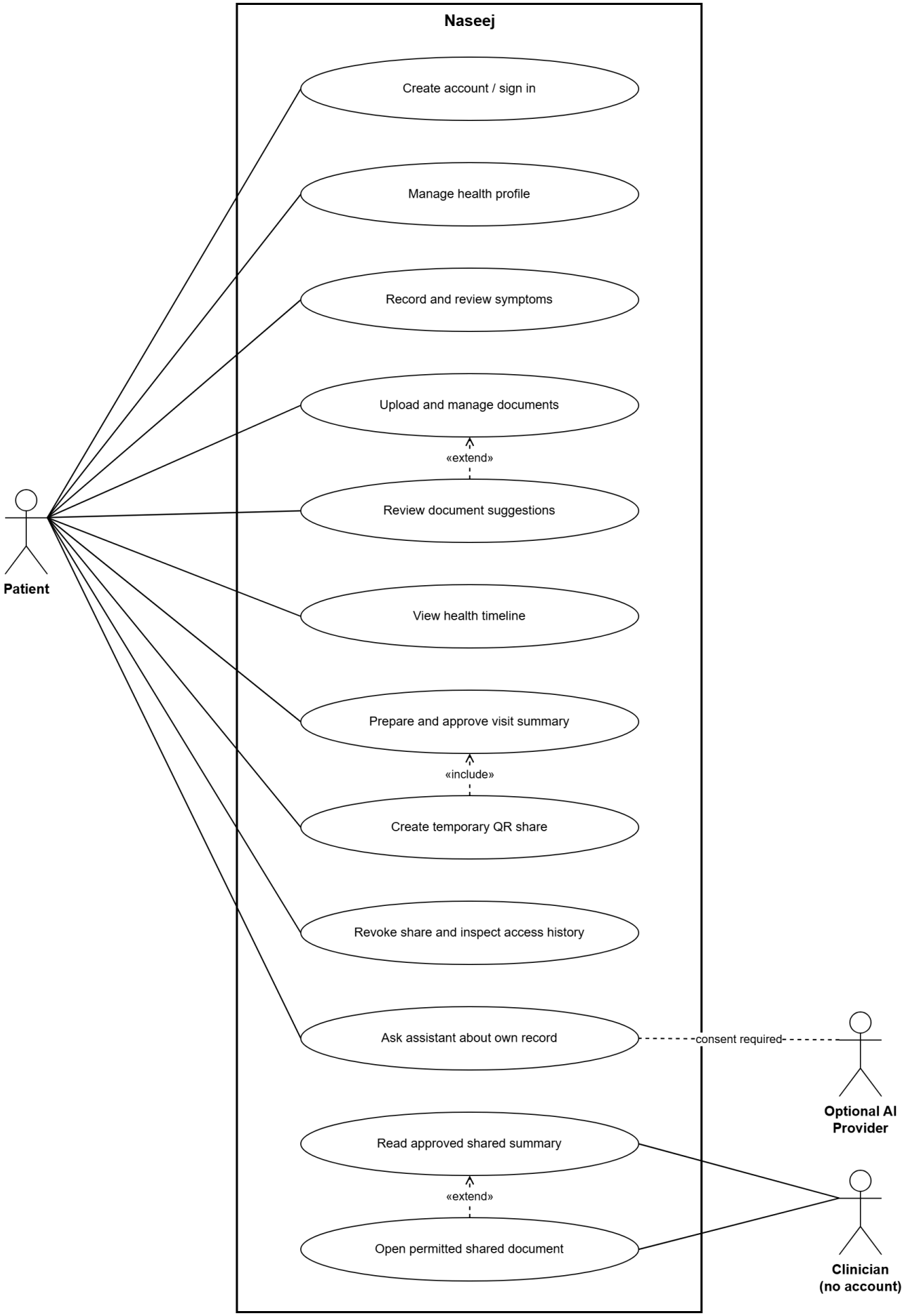
2. Flowchart: upload a document and review AI suggestions



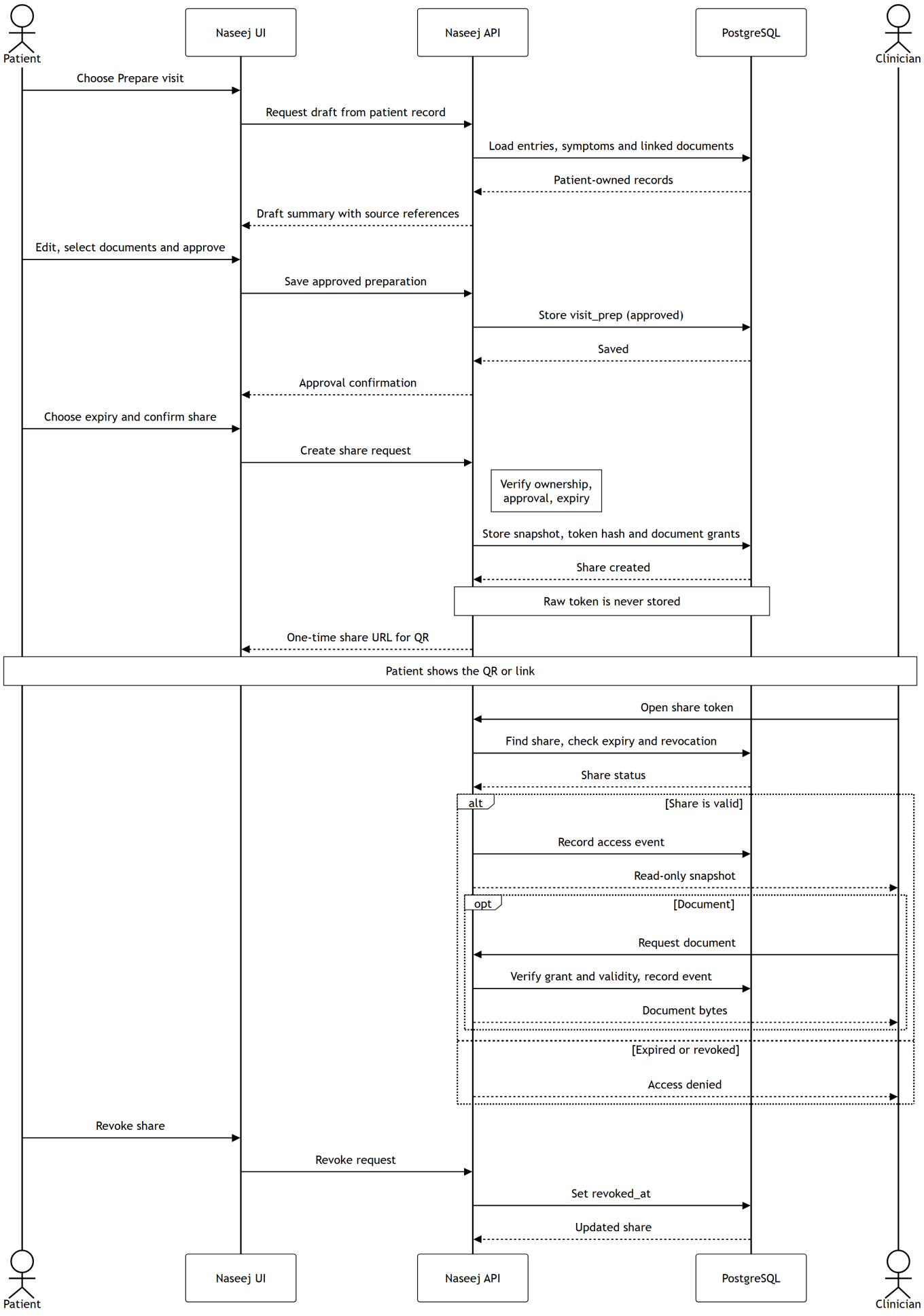
3. Activity diagram: prepare a visit summary and share it



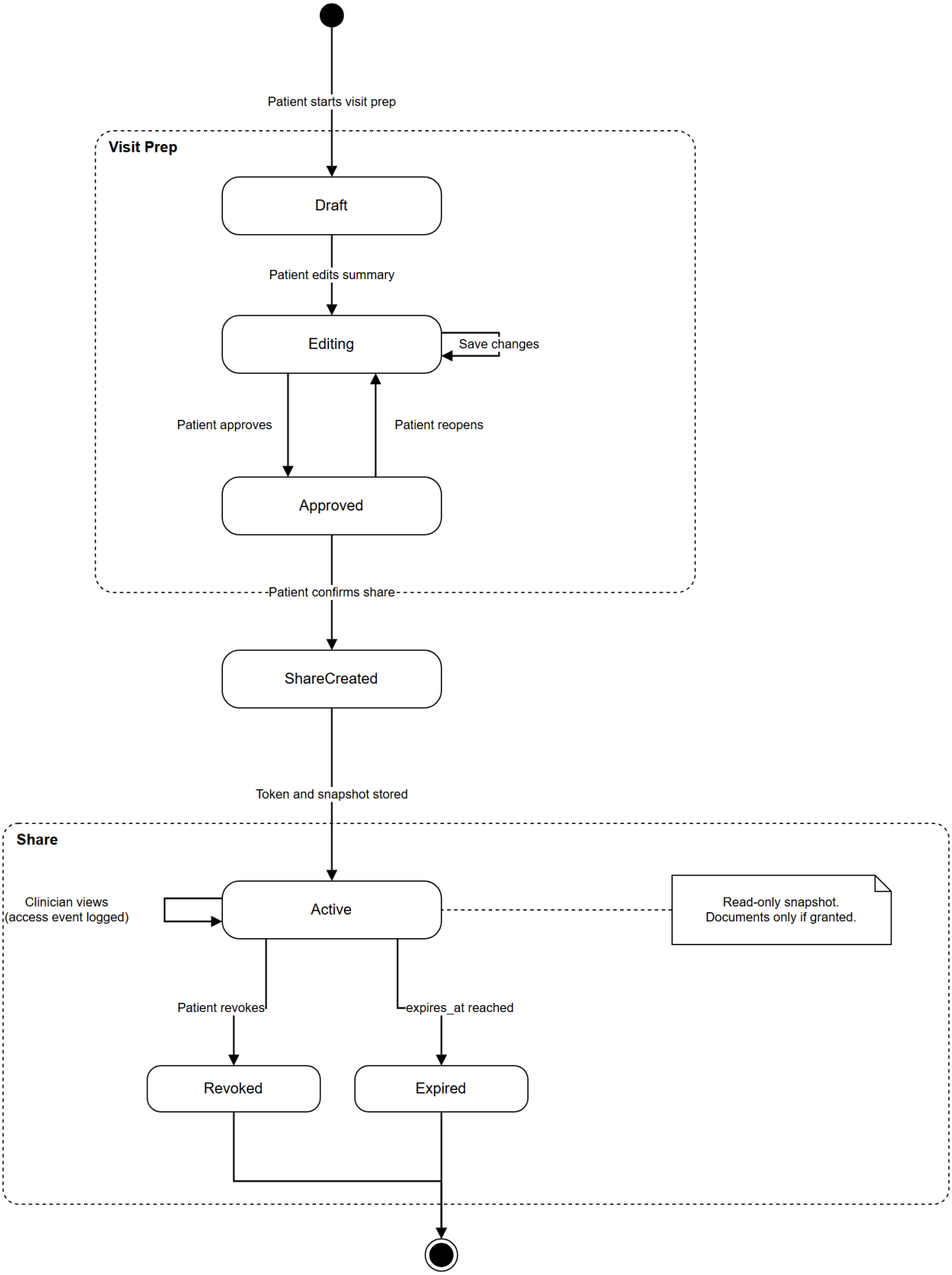
4. Use case diagram



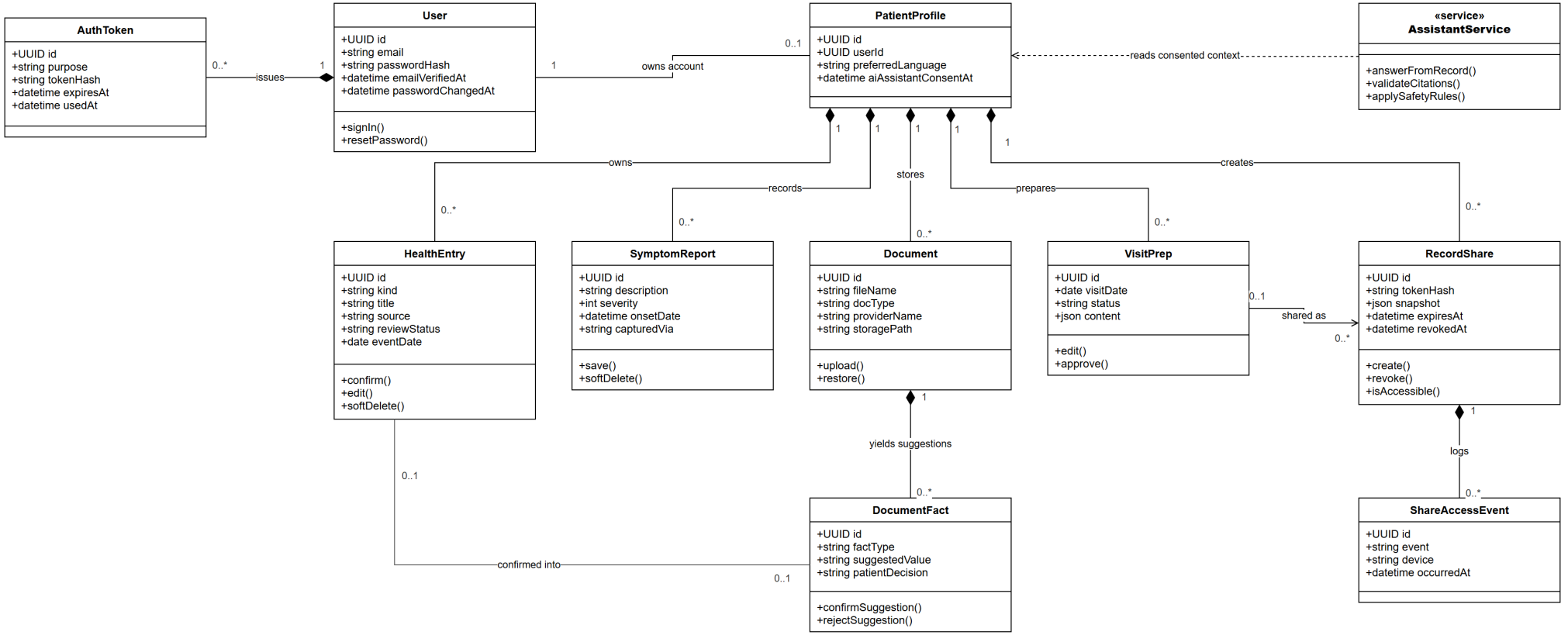
5. Sequence diagram: approve, share by QR, clinician reads, patient revokes



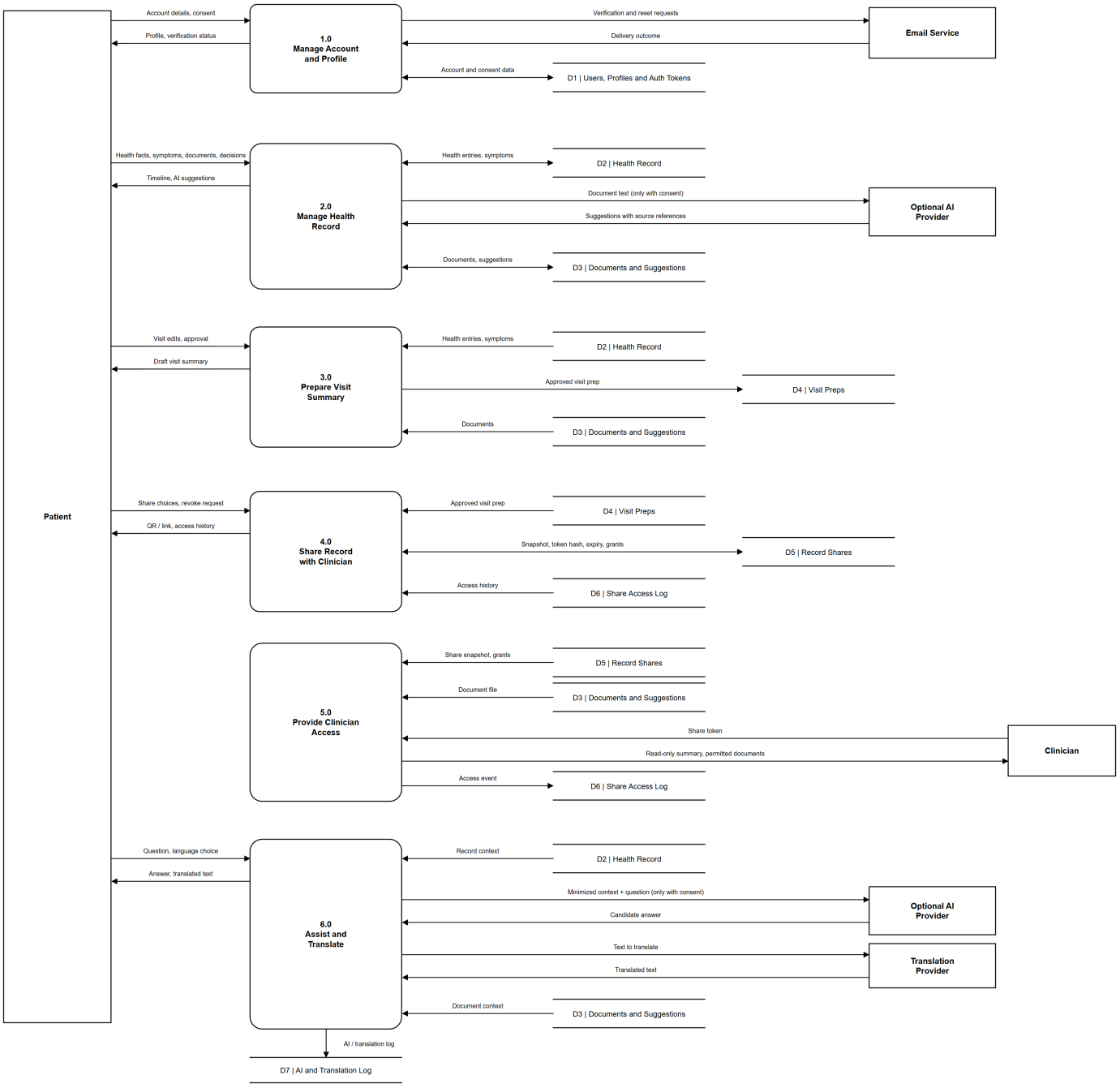
6. State diagram: visit preparation and share lifecycle



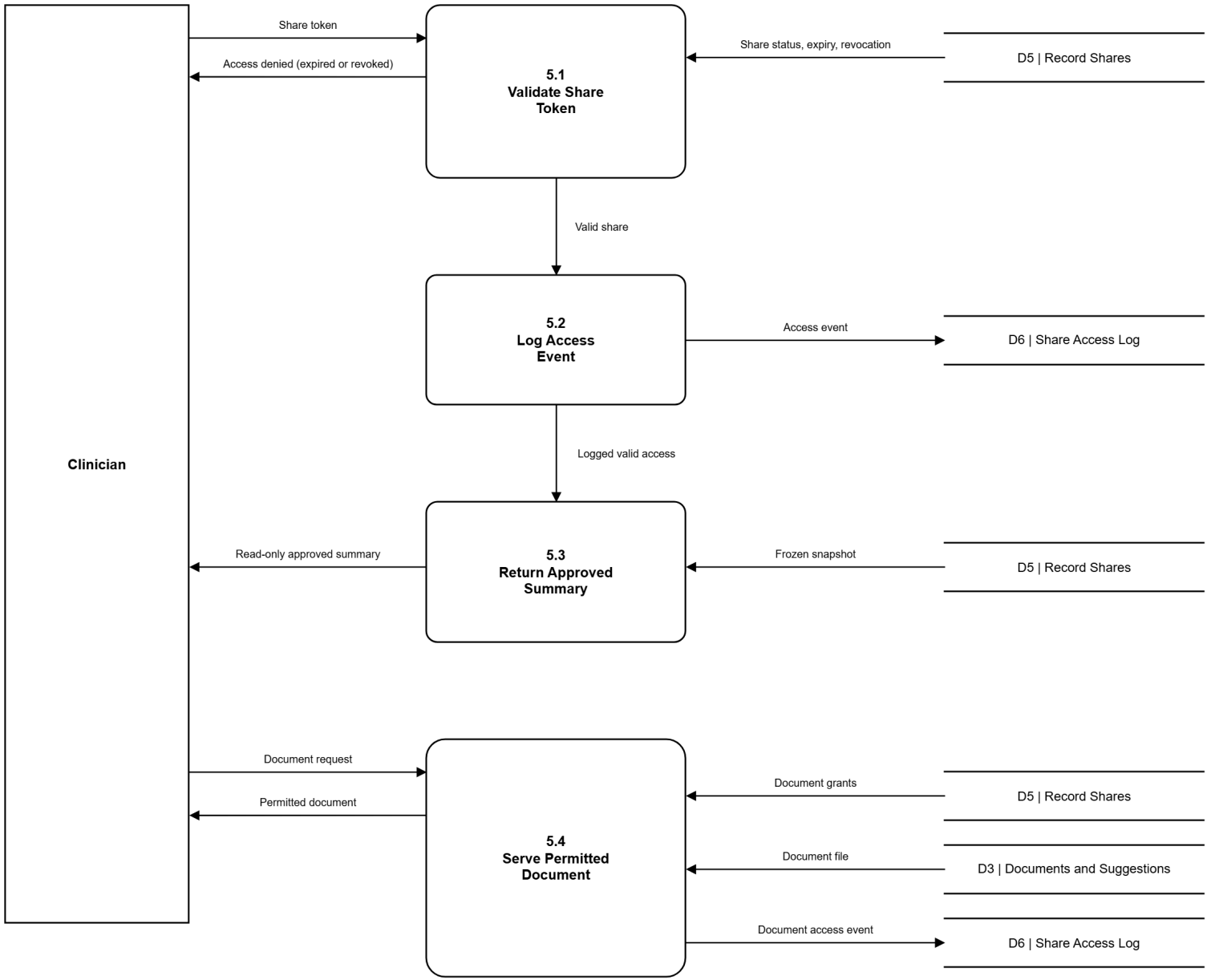
7. Class diagram



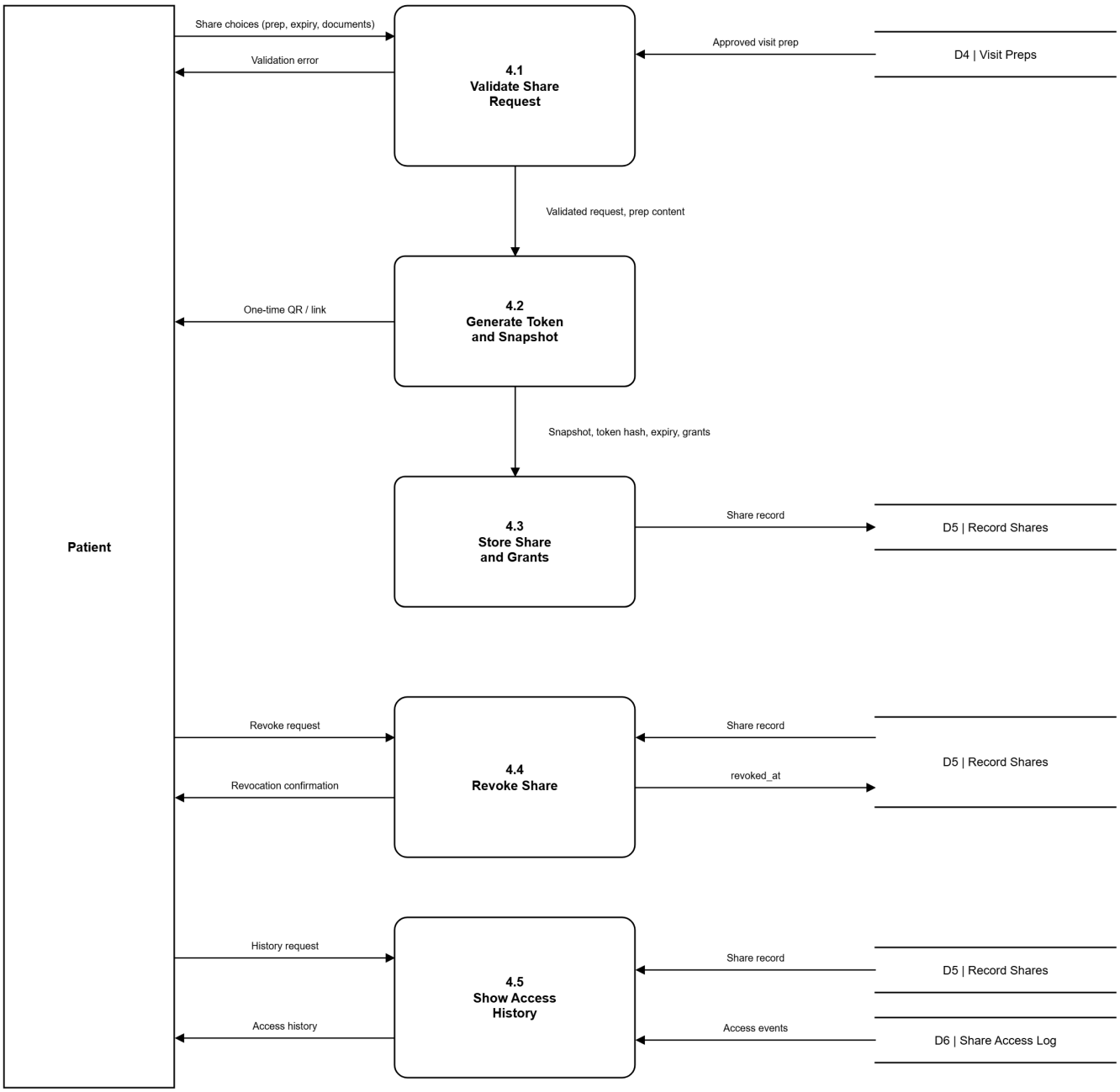
8. Data flow diagram, level 0



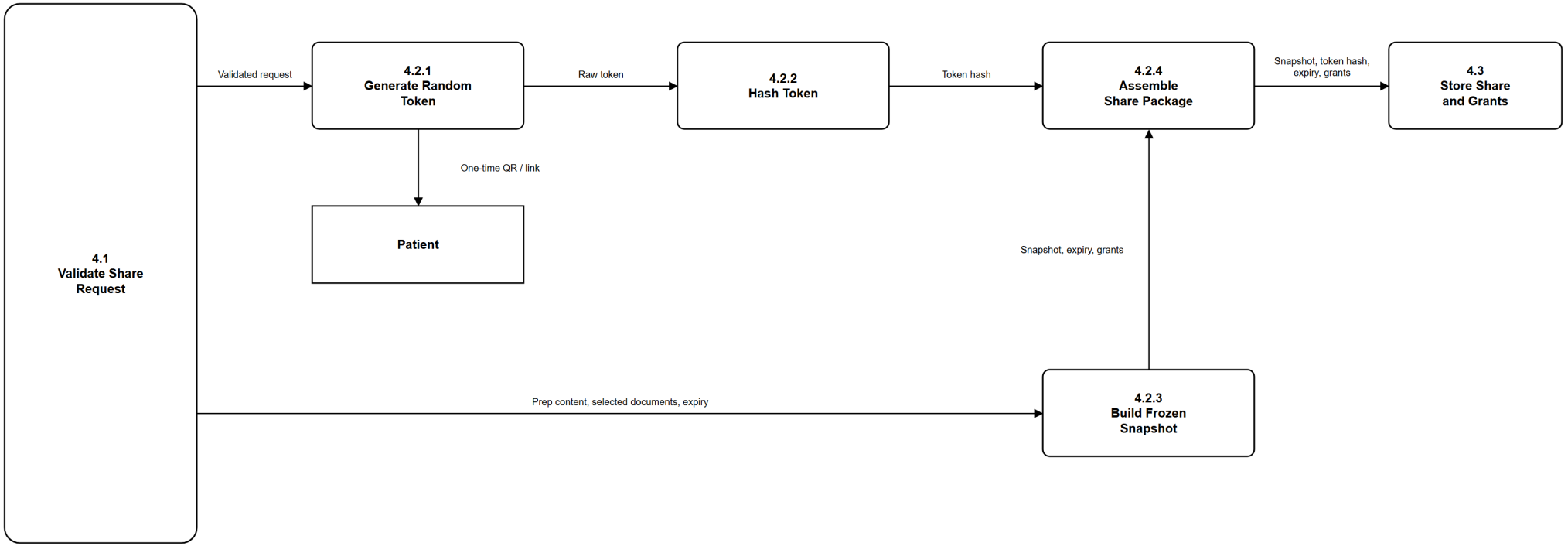
9. Data flow diagram, level 1: clinician access (5.0)



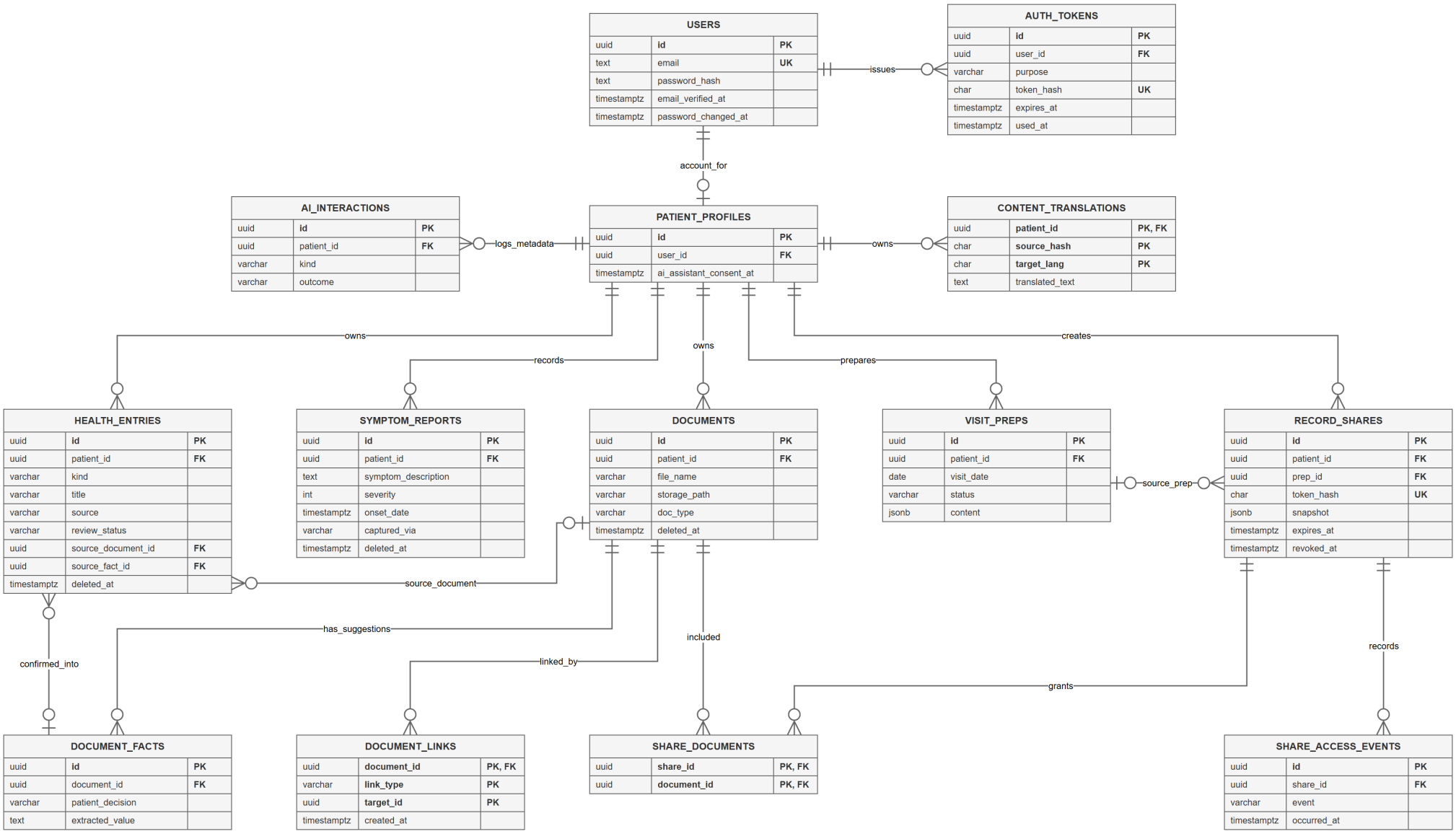
10. Data flow diagram, level 1: share record (4.0)



11. Data flow diagram, level 2: token and snapshot (4.2)



12. Entity relationship diagram (ERD)



ERD to Relational Schema Mapping

Naseej patient health record platform

Source docs/diagrams/source/erd-diagram.drawio **Tables** 14 **Relationships** 17

1. Mapping rules applied

ERD construct	Relational rule
Entity (strong)	Becomes one table. The entity identifier becomes the primary key (PK).
Attribute	Becomes a column of the table; the data type is kept as drawn in the ERD.
Unique attribute	Becomes a column with a UNIQUE key (UK), e.g. USERS.email, RECORD_SHARES.token_hash, AUTH_TOKENS.token_hash.
One-to-many (1 : N)	The primary key of the 1-side is copied into the N-side table as a foreign key (FK).
One-to-one, optional (1 : 0..1)	As one-to-many, but the foreign key is also declared UNIQUE so at most one row can point to the parent.
Many-to-many (M : N)	Resolved by an associative table whose composite PK is made of the two foreign keys (SHARE_DOCUMENTS).
Dependent entity	The table takes the parent key as part of a composite PK (CONTENT_TRANSLATIONS, DOCUMENT_LINKS).
Optional participation (0..1)	The foreign key column accepts NULL (RECORD_SHARES.prep_id, HEALTH_ENTRIES.source_document_id, HEALTH_ENTRIES.source_fact_id).

2. Relational schema

Notation: **TABLE** (primary key, *foreign key*, *key that is both*, other columns). Only the columns shown in the ERD are listed.

USERS (id, email, password_hash, email_verified_at, password_changed_at)

AUTH_TOKENS (id, *user_id*, purpose, token_hash, expires_at, used_at)

PATIENT_PROFILES (id, *user_id*, ai_assistant_consent_at)

AI_INTERACTIONS (id, *patient_id*, kind, outcome)

CONTENT_TRANSLATIONS (*patient_id*, source_hash, target_lang, translated_text)

HEALTH_ENTRIES (id, *patient_id*, kind, title, source, review_status, *source_document_id*, *source_fact_id*, deleted_at)

SYMPTOM_REPORTS (id, *patient_id*, symptom_description, severity, onset_date, captured_via, deleted_at)

DOCUMENTS (id, *patient_id*, file_name, storage_path, doc_type, deleted_at)

DOCUMENT_FACTS (id, *document_id*, patient_decision, extracted_value)

DOCUMENT_LINKS (*document_id*, link_type, target_id, created_at)

VISIT_PREPS (id, *patient_id*, visit_date, status, content)

RECORD_SHARES (id, *patient_id*, *prep_id*, token_hash, snapshot, expires_at, revoked_at)

SHARE_DOCUMENTS (*share_id*, *document_id*)

SHARE_ACCESS_EVENTS (id, *share_id*, event, occurred_at)

3. Relationship mapping

Each relationship drawn in the ERD and how it is realised as a foreign key. Cardinality is read parent : child.

#	Parent	Child	Relationship	Cardinality	Mapping result
1	**USERS**	**PATIENT_PROFILES**	account_for	1 : 0..1	FK PATIENT_PROFILES.user_id → USERS.id, with a UNIQUE constraint
2	**USERS**	**AUTH_TOKENS**	issues	1 : 0..*	FK AUTH_TOKENS.user_id → USERS.id
3	**PATIENT_PROFILES**	**AI_INTERACTIONS**	logs_metadata	1 : 0..*	FK AI_INTERACTIONS.patient_id → PATIENT_PROFILES.id
4	**PATIENT_PROFILES**	**CONTENT_TRANSLATIONS**	owns	1 : 0..*	FK is part of the composite PK (patient_id, source_hash, target_lang)
5	**PATIENT_PROFILES**	**HEALTH_ENTRIES**	owns	1 : 0..*	FK HEALTH_ENTRIES.patient_id → PATIENT_PROFILES.id
6	**PATIENT_PROFILES**	**SYMPTOM_REPORTS**	records	1 : 0..*	FK SYMPTOM_REPORTS.patient_id → PATIENT_PROFILES.id
7	**PATIENT_PROFILES**	**DOCUMENTS**	owns	1 : 0..*	FK DOCUMENTS.patient_id → PATIENT_PROFILES.id
8	**PATIENT_PROFILES**	**VISIT_PREPS**	prepares	1 : 0..*	FK VISIT_PREPS.patient_id → PATIENT_PROFILES.id
9	**PATIENT_PROFILES**	**RECORD_SHARES**	creates	1 : 0..*	FK RECORD_SHARES.patient_id → PATIENT_PROFILES.id
10	**VISIT_PREPS**	**RECORD_SHARES**	source_prep	0..1 : 0..*	Optional FK RECORD_SHARES.prep_id → VISIT_PREPS.id (nullable)
11	**DOCUMENTS**	**DOCUMENT_FACTS**	has_suggestions	1 : 0..*	FK DOCUMENT_FACTS.document_id → DOCUMENTS.id
12	**DOCUMENTS**	**HEALTH_ENTRIES**	source_document	0..1 : 0..*	Optional FK HEALTH_ENTRIES.source_document_id → DOCUMENTS.id (nullable)
13	**DOCUMENT_FACTS**	**HEALTH_ENTRIES**	confirmed_into	0..1 : 0..*	Optional FK HEALTH_ENTRIES.source_fact_id → DOCUMENT_FACTS.id (nullable)
14	**DOCUMENTS**	**DOCUMENT_LINKS**	linked_by	1 : 0..*	FK is part of the composite PK (document_id, link_type, target_id)
15	**DOCUMENTS**	**SHARE_DOCUMENTS**	included	1 : 0..*	M:N resolved: FK is part of the composite PK (share_id, document_id)
16	**RECORD_SHARES**	**SHARE_DOCUMENTS**	grants	1 : 0..*	M:N resolved: FK is part of the composite PK (share_id, document_id)
17	**RECORD_SHARES**	**SHARE_ACCESS_EVENTS**	records	1 : 0..*	FK SHARE_ACCESS_EVENTS.share_id → RECORD_SHARES.id

4. Table specifications

PK = primary key, FK = foreign key, UK = unique key. Data types are those shown in the ERD.

USERS

System account (credentials, verification and password changes)

Column	Data type	Key	References
id	uuid	PK	
email	text	UK	
password_hash	text		
email_verified_at	timestamptz		
password_changed_at	timestamptz		

AUTH_TOKENS

Single-use email verification and password reset links

Column	Data type	Key	References
id	uuid	PK	
user_id	uuid	FK	USERS.id
purpose	varchar		
token_hash	char	UK	
expires_at	timestamptz		
used_at	timestamptz		

PATIENT_PROFILES

Patient identity; owner of every record below

Column	Data type	Key	References
id	uuid	PK	
user_id	uuid	FK, UK	USERS.id
ai_assistant_consent_at	timestamptz		

AI_INTERACTIONS

Metadata log of assistant calls

Column	Data type	Key	References
id	uuid	PK	
patient_id	uuid	FK	PATIENT_PROFILES.id
kind	varchar		
outcome	varchar		

CONTENT_TRANSLATIONS

Cached translation per patient, source and language

Column	Data type	Key	References
patient_id	uuid	PK, FK	PATIENT_PROFILES.id
source_hash	char	PK	
target_lang	char	PK	
translated_text	text		

HEALTH_ENTRIES

Conditions, medications, allergies, measurements

Column	Data type	Key	References
id	uuid	PK	
patient_id	uuid	FK	PATIENT_PROFILES.id
kind	varchar		
title	varchar		
source	varchar		
review_status	varchar		
source_document_id	uuid	FK	DOCUMENTS.id
source_fact_id	uuid	FK	DOCUMENT_FACTS.id
deleted_at	timestamptz		

SYMPTOM_REPORTS

Patient-logged symptoms

Column	Data type	Key	References
id	uuid	PK	
patient_id	uuid	FK	PATIENT_PROFILES.id
symptom_description	text		
severity	int		
onset_date	timestamptz		
captured_via	varchar		
deleted_at	timestamptz		

DOCUMENTS

Uploaded files kept in the patient vault

Column	Data type	Key	References
id	uuid	PK	
patient_id	uuid	FK	PATIENT_PROFILES.id
file_name	varchar		
storage_path	varchar		
doc_type	varchar		
deleted_at	timestamptz		

DOCUMENT_FACTS

Suggestions extracted from a document

Column	Data type	Key	References
id	uuid	PK	
document_id	uuid	FK	DOCUMENTS.id
patient_decision	varchar		
extracted_value	text		

DOCUMENT_LINKS

Links a document to a symptom, entry or visit

Column	Data type	Key	References
document_id	uuid	PK, FK	DOCUMENTS.id
link_type	varchar	PK	
target_id	uuid	PK	(polymorphic, see note 3)
created_at	timestamptz		

VISIT_PREPS

Visit preparation sheet

Column	Data type	Key	References
id	uuid	PK	
patient_id	uuid	FK	PATIENT_PROFILES.id
visit_date	date		
status	varchar		
content	jsonb		

RECORD_SHARES

Time-limited share of approved content

Column	Data type	Key	References
id	uuid	PK	
patient_id	uuid	FK	PATIENT_PROFILES.id
prep_id	uuid	FK	VISIT_PREPS.id
token_hash	char	UK	
snapshot	jsonb		
expires_at	timestamptz		
revoked_at	timestamptz		

SHARE_DOCUMENTS

Documents a share may stream

Column	Data type	Key	References
share_id	uuid	PK, FK	RECORD_SHARES.id
document_id	uuid	PK, FK	DOCUMENTS.id

SHARE_ACCESS_EVENTS

Append-only access log of a share

Column	Data type	Key	References
id	uuid	PK	
share_id	uuid	FK	RECORD_SHARES.id
event	varchar		
occurred_at	timestamptz		

5. Mapping notes

1. **Ownership.** Every record table reaches PATIENT_PROFILES either directly through patient_id or through its parent (DOCUMENT_FACTS and DOCUMENT_LINKS through DOCUMENTS; SHARE_DOCUMENTS and SHARE_ACCESS_EVENTS through RECORD_SHARES). USERS and AUTH_TOKENS belong to the account, not to the profile.
2. **Provenance foreign keys.** HEALTH_ENTRIES.source_document_id and HEALTH_ENTRIES.source_fact_id reference DOCUMENTS and DOCUMENT_FACTS. They record where an entry came from, are optional (NULL for an entry the patient typed), and are drawn as the source_document and confirmed_into relationships.
3. **Polymorphic key.** DOCUMENT_LINKS.target_id points to a symptom report, a health entry or a visit depending on link_type. A single database foreign key cannot express this, so it is part of the PK without an FK constraint and the application checks it.
4. **Profile to account.** PATIENT_PROFILES.user_id is UNIQUE because the ERD allows at most one profile per account (1 : 0..1).
5. **Soft delete.** HEALTH_ENTRIES, SYMPTOM_REPORTS and DOCUMENTS keep deleted_at instead of removing rows.
6. **Auth tokens.** AUTH_TOKENS (migration 007) holds the links mailed for email verification and password reset. Only the SHA-256 of a token is stored (token_hash, UNIQUE); a token is single-use (used_at) and expires (expires_at). USERS.password_changed_at invalidates sessions issued before a reset. Rows are deleted with their user (ON DELETE CASCADE).